11. Sequences and Series

11.5. Alternating Series and Absolute Convergence

11.5 Learning Objectives
- Identify alternating series
- Determine if alternating series converge or diverge by the alternating series test
- Estimate the reaminder for alternating series
- Distinguish between absolute and conditional convergence

The convergence tests we have looked at so far only apply to series with *positive terms*. In this section, we will focus on *alternating series*.

Definition 11.5.1

A series $\sum_{n=1}^{\infty} a_n$ is an alternating series if the series terms have the form

$$a_n = (-1)^n b_n \quad \text{or} \quad a_n = (-1)^{n-1} b_n \quad \text{for} \quad b_n \geq 0.$$

Example 11.5.1

Is the following series and alternating series or not?

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}(n+1)}{n+2}$$

Solution

Yes, this series is alternating because $b_n = \frac{n+1}{n+2} \geq 0$ for all $n \geq 1$, and the $(-1)^{n-1}$ in front make the terms alternate between positive and negative.

Example 11.5.2

Is the following series and alternating series or not?

$$\sum_{n=1}^{\infty} \frac{\cos(\pi n)}{\sqrt{n}}$$

Solution

Yes, for integers n, we can rewrite $\cos(\pi n) = (-1)^n$ and $\frac{1}{\sqrt{n}} \geq 0$ for $n \geq 1$.

Example 11.5.3

Is the following series and alternating series or not?

$$\sum_{n=1}^{\infty} \frac{\cos(2\pi n)}{n^2}$$

Solution

No, because for integer n, $\cos(2\pi n) = 1$ and so this series is the same as $\sum_{n=1}^{\infty} \frac{1}{n^2}$ and $\frac{1}{n^2} \geq 0$.

Example 11.5.4

Is the following series and alternating series or not?

$$\sum_{n=1}^{\infty} (-1)^n \cos(n)$$

Solution

No, while there is the $(-1)^n$ part, $b_n = \cos(n) \not\geq 0$, in other words $\cos(n)$ is positive or negative for different values of $n \geq 1$ and we need $b_n \geq 0$.

Theorem 11.5.1 Alternating Series Test

If an alternating series, $\sum_{n=1}^{\infty} a_n$ of the form

$$\sum_{n=1}^{\infty} (-1)^{n-1} b_n \quad \text{or} \quad \sum_{n=1}^{\infty} (-1)^n b_n,$$

where $b_n \geq 0$ satisfies:

(i) $b_{n+1} \leq b_n$ for all $n \geq N$ (N is some number, this requirement is that ""b_n is decreasing eventually")

(ii) $\lim_{n\to\infty} b_n = 0$

then the series $\sum_{n=1}^{\infty} a_n$ is convergent.

Warning

A common mistake students make at this point is to look at *any* series and say that the terms are decreasing to 0 so the series converges, however we know from the divergence test (section 11.2b) that this does not guarantee the series converges. This is only the case in the very special **alternating** series test.

The reasoning for why the alternating series test works is because the terms are decreasing in absolute value but alternating sign, we can always verify that the sequence of partial sums converges.

todo: picture

Example 11.5.5

Prove that the **alternating harmonic series**

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$$

converges.

Solution

Notice that $b_n = \frac{1}{n}$ is decreasing since

$$n+1 > n \quad \text{for} \quad n \geq 1$$
$$\frac{1}{n+1} < \frac{1}{n}$$
$$b_{n+1} < b_n$$

Also notice that

$$\lim_{n\to\infty} b_n = \lim_{n\to\infty} \frac{1}{n} = 0.$$

Therefore, by the alternating series test,

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$$

converges.

Recall that the (regular) harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges, but the alternating series converges. This shows that alternating series are more likely to converge.

Example 11.5.6

Determine whether

$$\sum_{n=2}^{\infty} \frac{(-1)^{n+1} \ln(n)}{n}$$

is convergent or divergent.

Solution

This is an alternating series with $b_n = \frac{\ln(n)}{n}$. First, notice that

$$\lim_{n\to\infty} b_n = \lim_{n\to\infty} \frac{\ln(n)}{n} \overset{\text{LH}}{=} \lim_{n\to\infty} \frac{\frac{1}{n}}{1} = 0.$$

Then, we need to show that

$$\frac{\ln(n)}{n}$$

is decreasing eventually. It is difficult to argue clearly why $\frac{\ln(n)}{n} > \frac{\ln(n+1)}{n+1}$, but we can use something else in calculus to show when something is decreasing...derivatives!

Let $y = \frac{\ln(x)}{x}$. Then, by quotient rule,

$$\begin{aligned} y' &= \frac{x \cdot \frac{1}{x} - \ln(x)}{x^2} \\ &= \frac{1 - \ln(x)}{x^2} \end{aligned}$$

To find the critical points, we set $y' = 0$ and find where y' does not exist. Then, these points are $x = e, 0$. On (e, ∞) y' is negative, so after $x = e$, y is decreasing. Going back to integers, for $n \geq 3$, $\frac{\ln(n)}{n}$ is decreasing ($N = 3$). Therefore, by the alternating series test, the series

$$\sum_{n=2}^{\infty} \frac{(-1)^{n+1} \ln(n)}{n}$$

converges.

Example 11.5.7

Determine whether the series

$$\sum_{n=1}^{\infty} \frac{(-1)^n n}{n+1}$$

is convergent or divergent.

Solution

Observe that the alternating series test cannot be applied here since $\frac{n}{n+1}$ is *increasing* for $n \geq 1$, not decreasing. We can see this by noticing

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{x}{x+1}\right) = \frac{x+1-x}{(x+1)^2} = \frac{1}{(x+1)^2} > 0$$

for all x.

However, this is a good time to remember we should be checking the divergence test at the start of these problems, notice

$$\lim_{n\to\infty} \frac{(-1)^n n}{n+1}$$

does not exist (the odd n terms go to -1 and the even n terms go to 1). So, the series

$$\sum_{n=1}^{\infty} \frac{(-1)^n n}{n+1}$$

diverges by the divergence test.

11.4a Section Summary:

- We looked at identifying alternating series and applying the alternating series test.